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(54) **STATIC RANDOM ACCESS MEMORY WITH
MAGNETIC TUNNEL JUNCTION CELLS**

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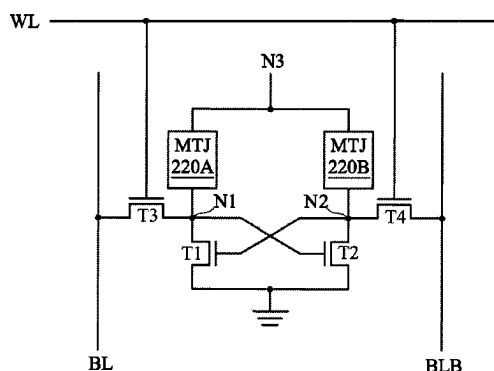
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ABSTRACT

Disclosed herein are related to a memory cell including magnetic tunneling junction (MTJ) devices. In one aspect, the memory cell includes a first layer including a first transistor and a second transistor. In one aspect, the first transistor and the second transistor are connected to each other in a cross-coupled configuration. A first drain structure of the first transistor may be electrically coupled to a first gate structure of the second transistor, and a second drain structure of the second transistor may be electrically coupled to a second gate structure of the first transistor. In one aspect, the memory cell includes a second layer including a first MTJ device electrically coupled to the first drain structure of the first transistor and a second MTJ device electrically coupled to the second drain structure of the second transistor. In one aspect, the second layer is above the first layer.

20 Claims, 13 Drawing Sheets

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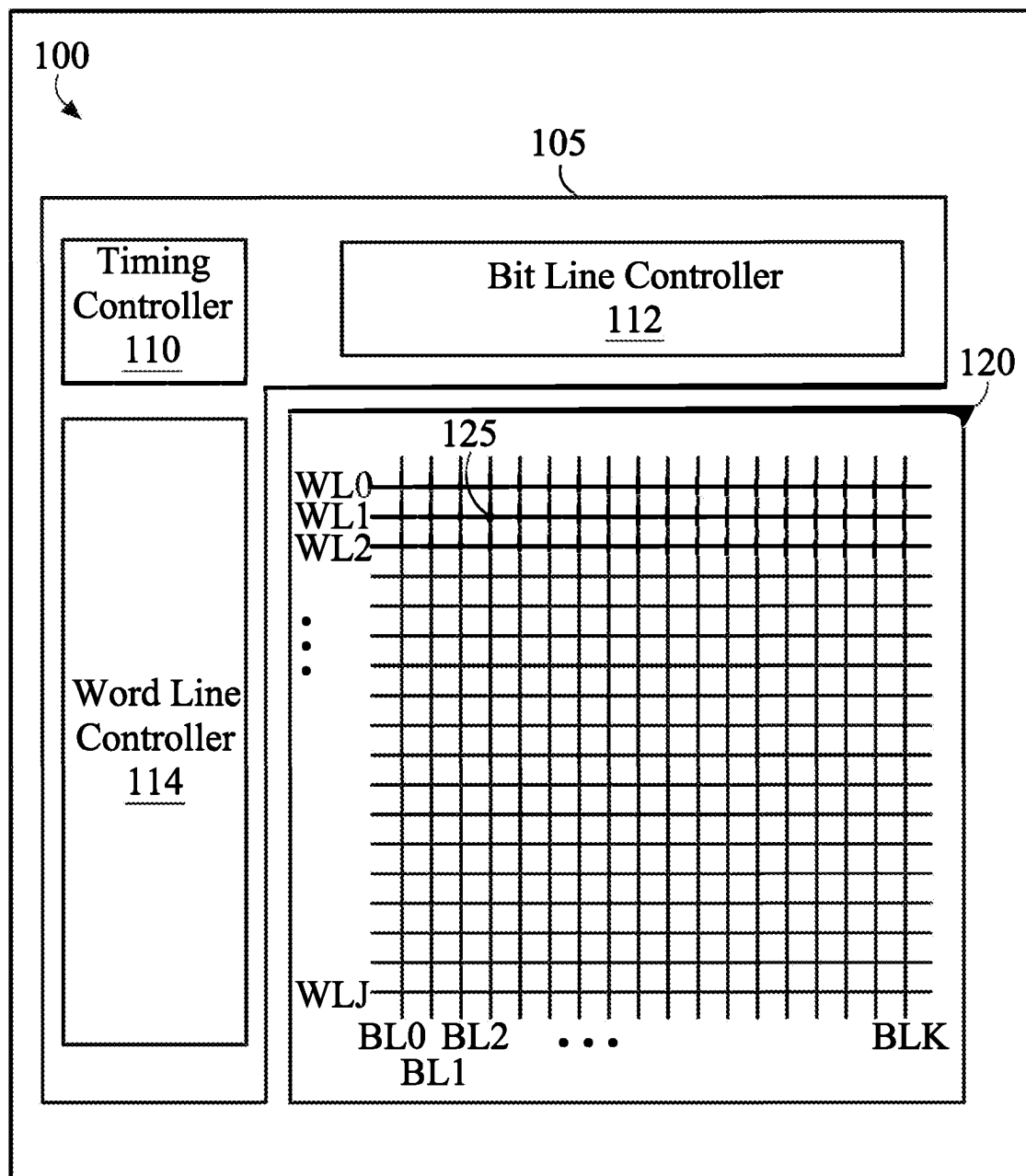


FIG.1

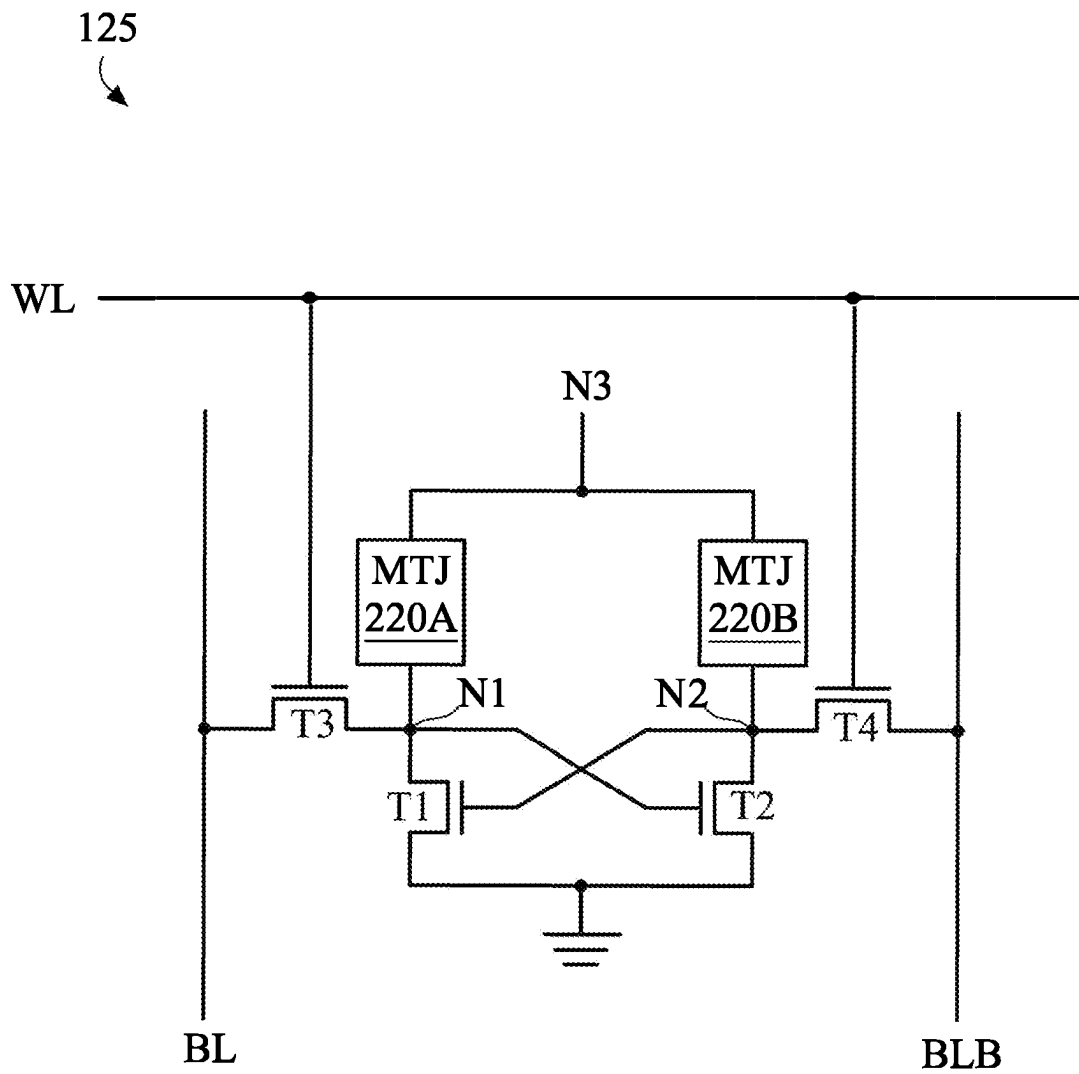
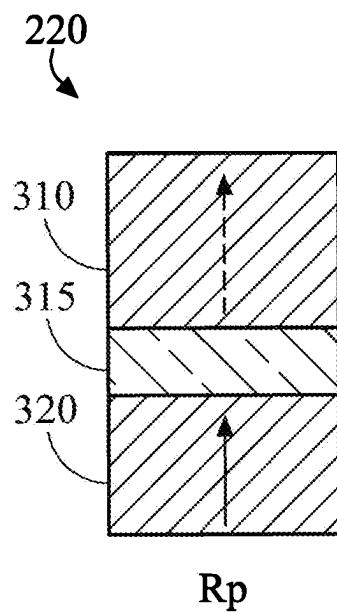


FIG. 2

Parallel Magnetic State



Anti-Parallel Magnetic State

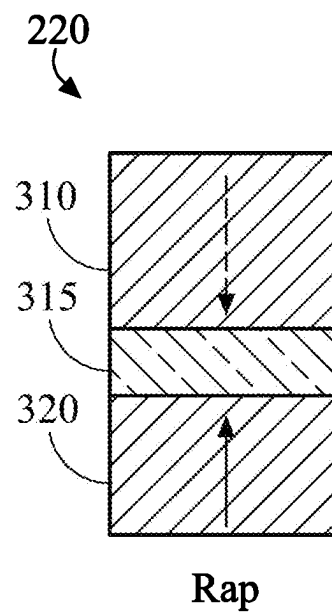


FIG. 3A

FIG. 3B

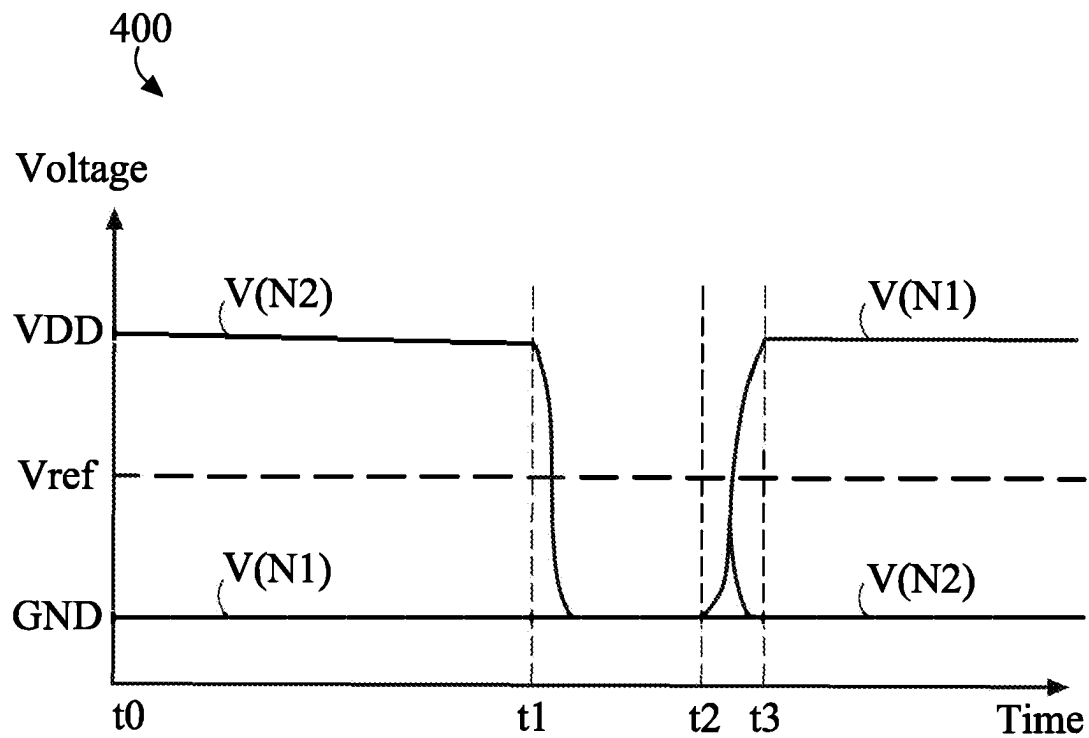


FIG. 4

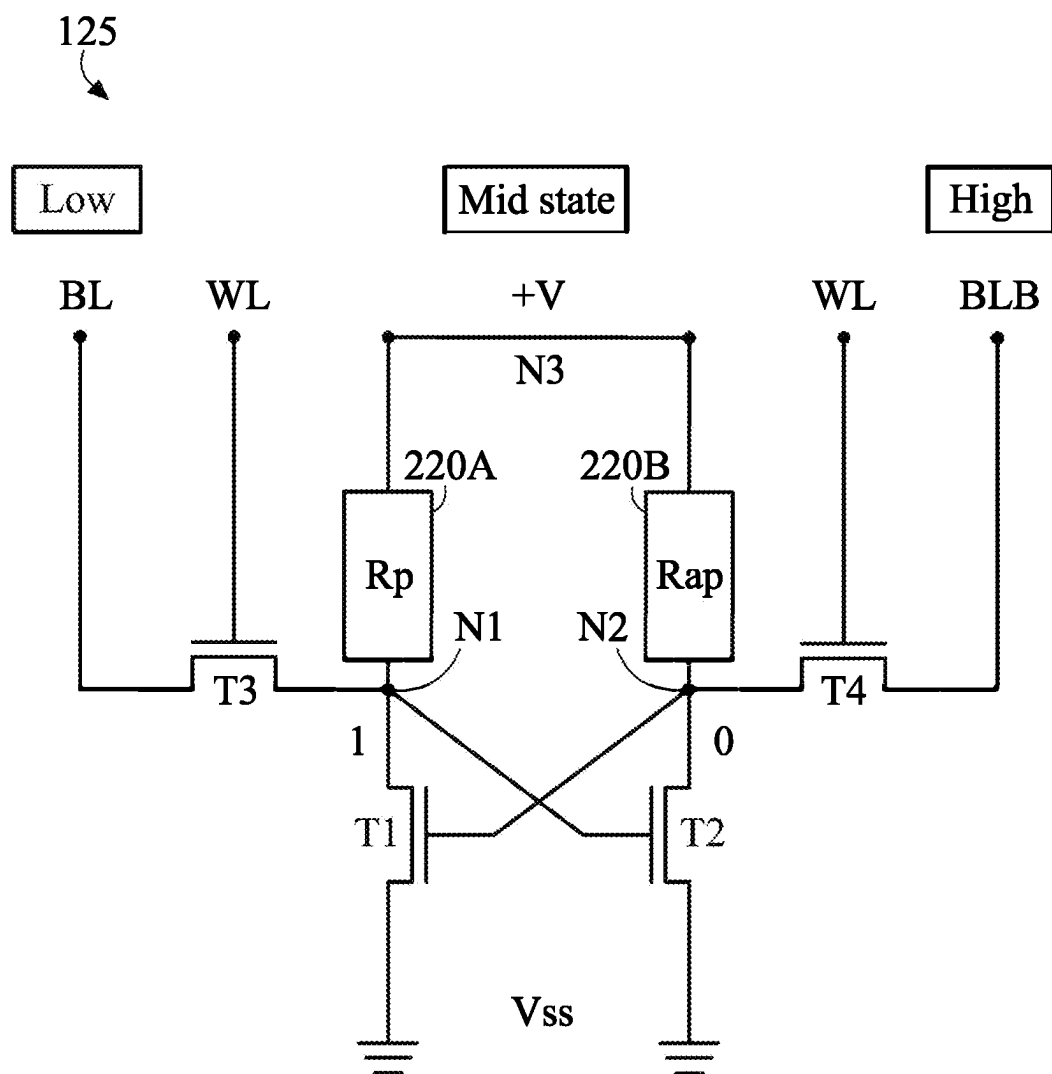


FIG. 5A

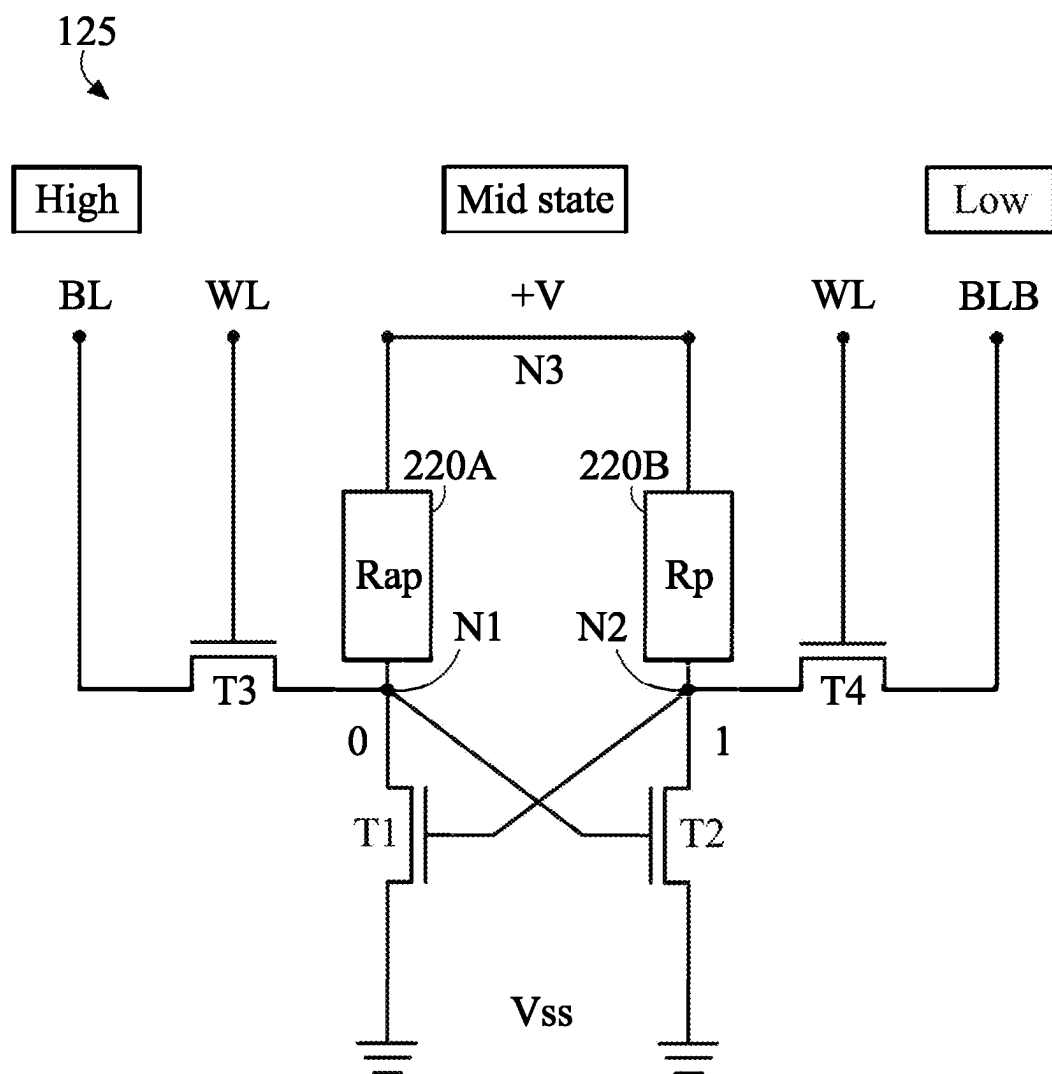


FIG. 5B

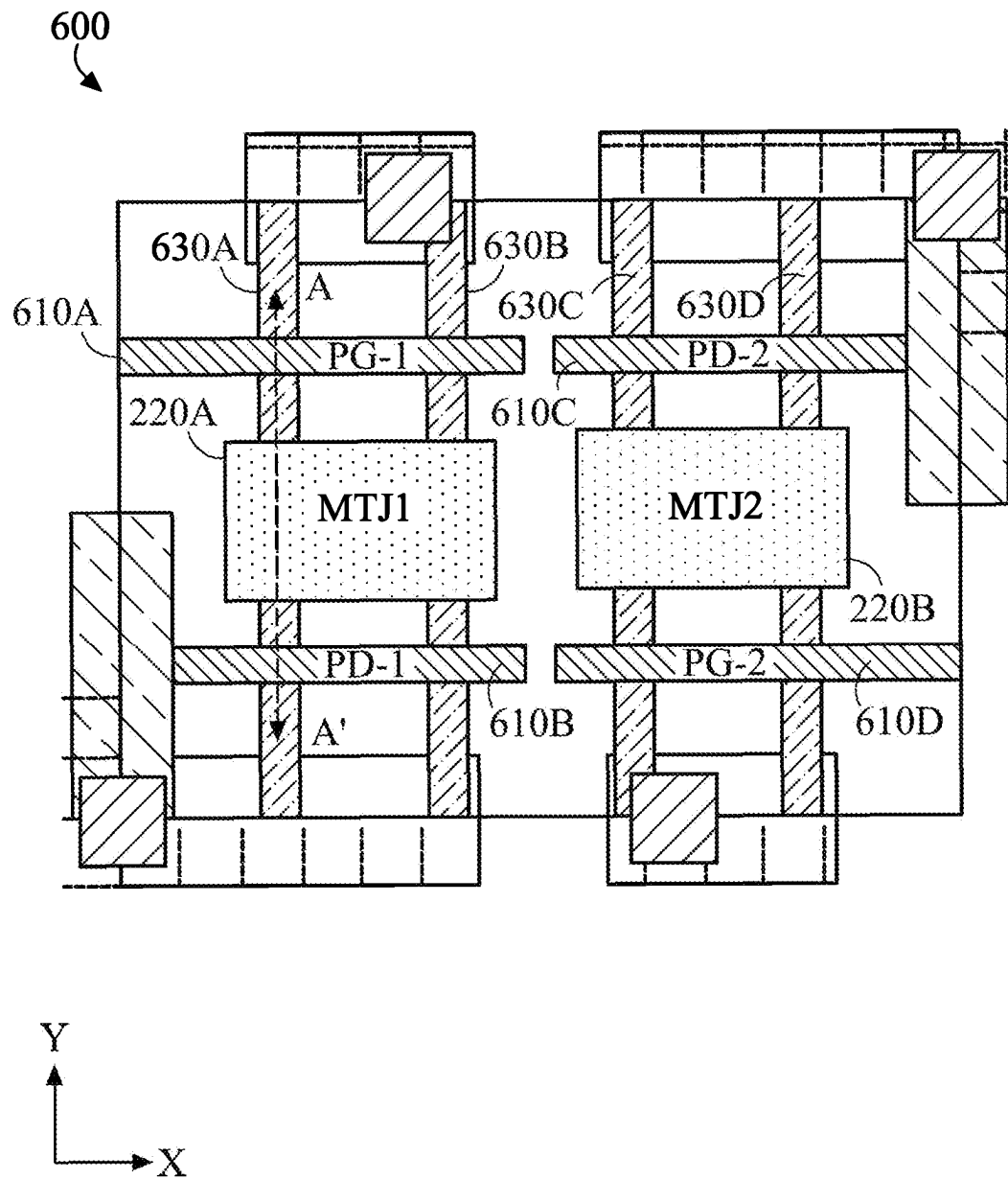


FIG. 6

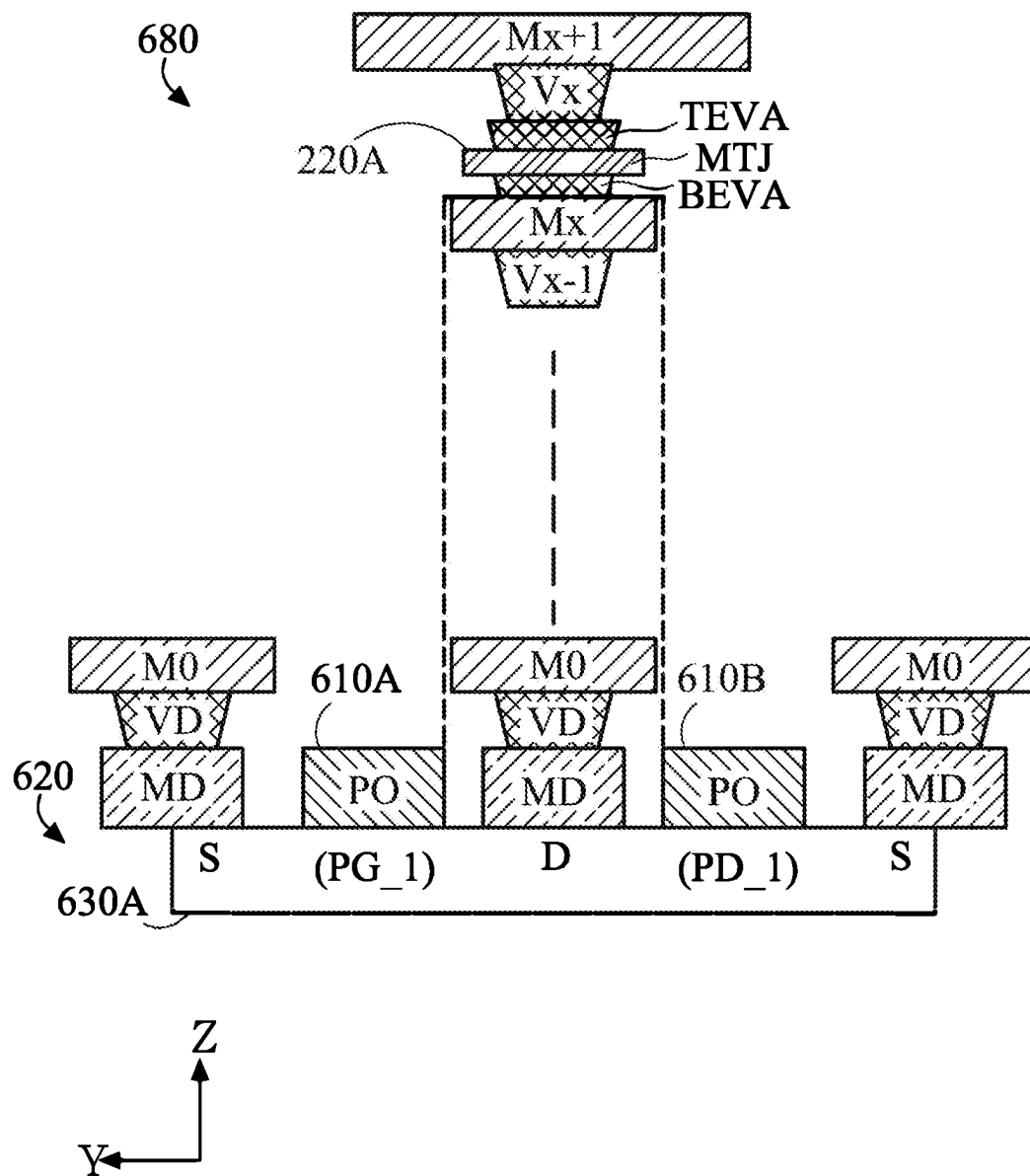


FIG. 7

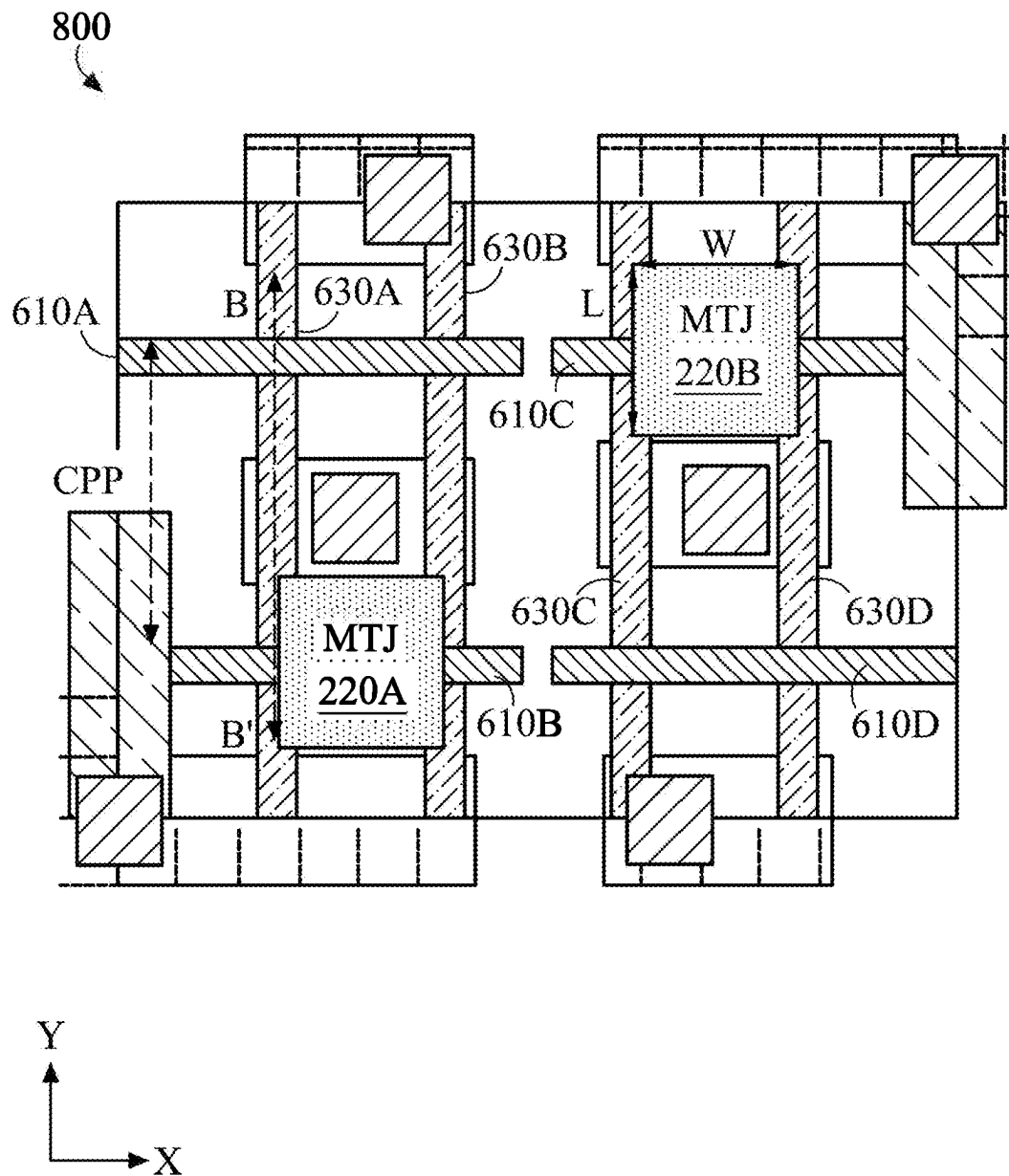


FIG. 8

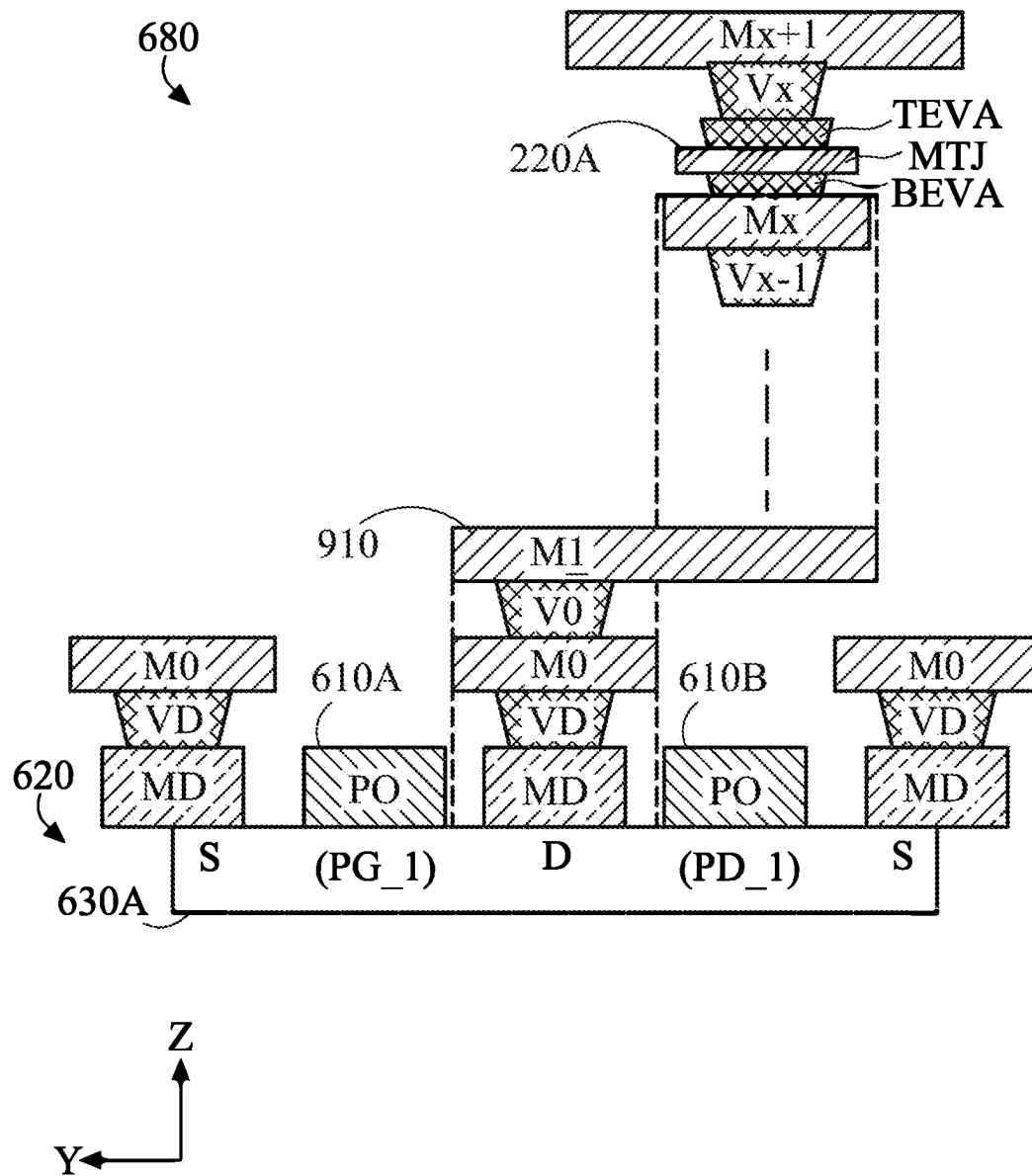
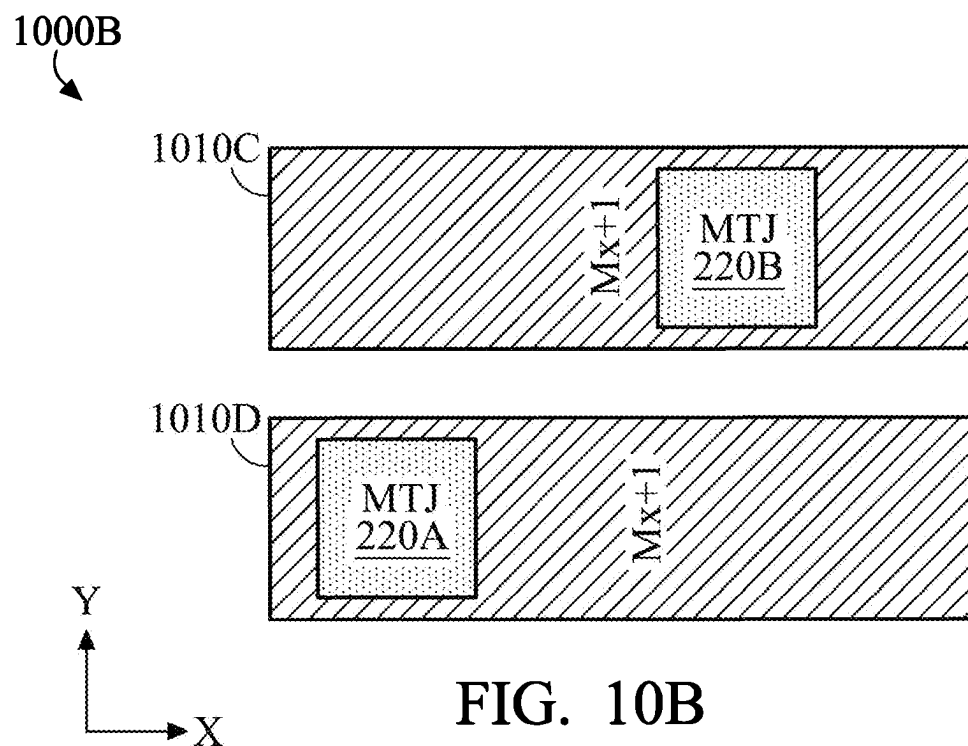
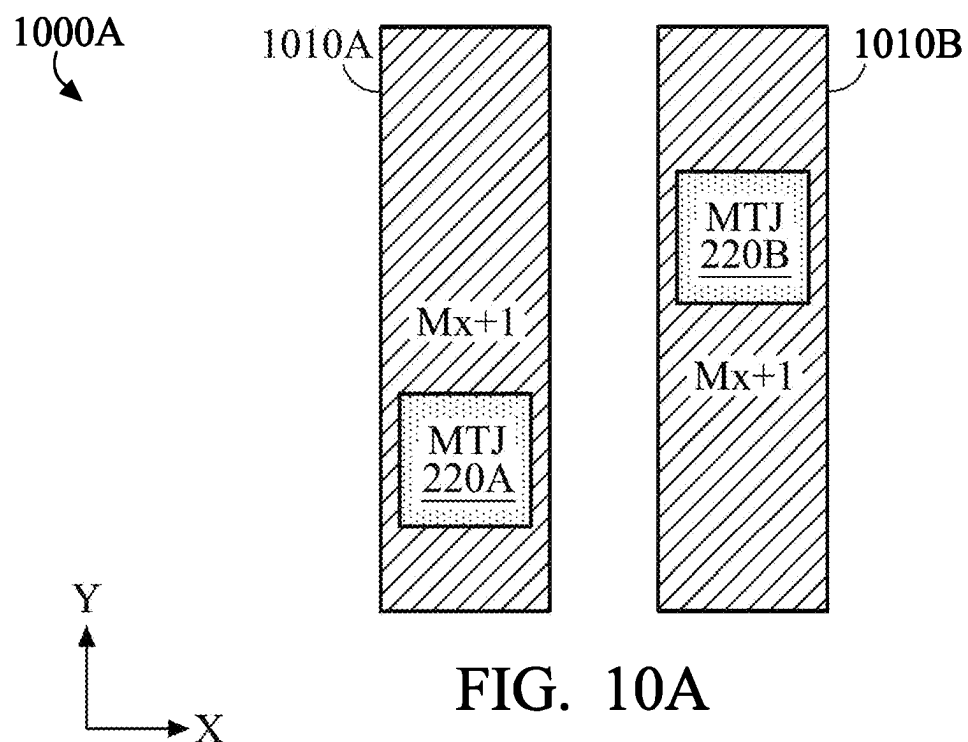


FIG. 9



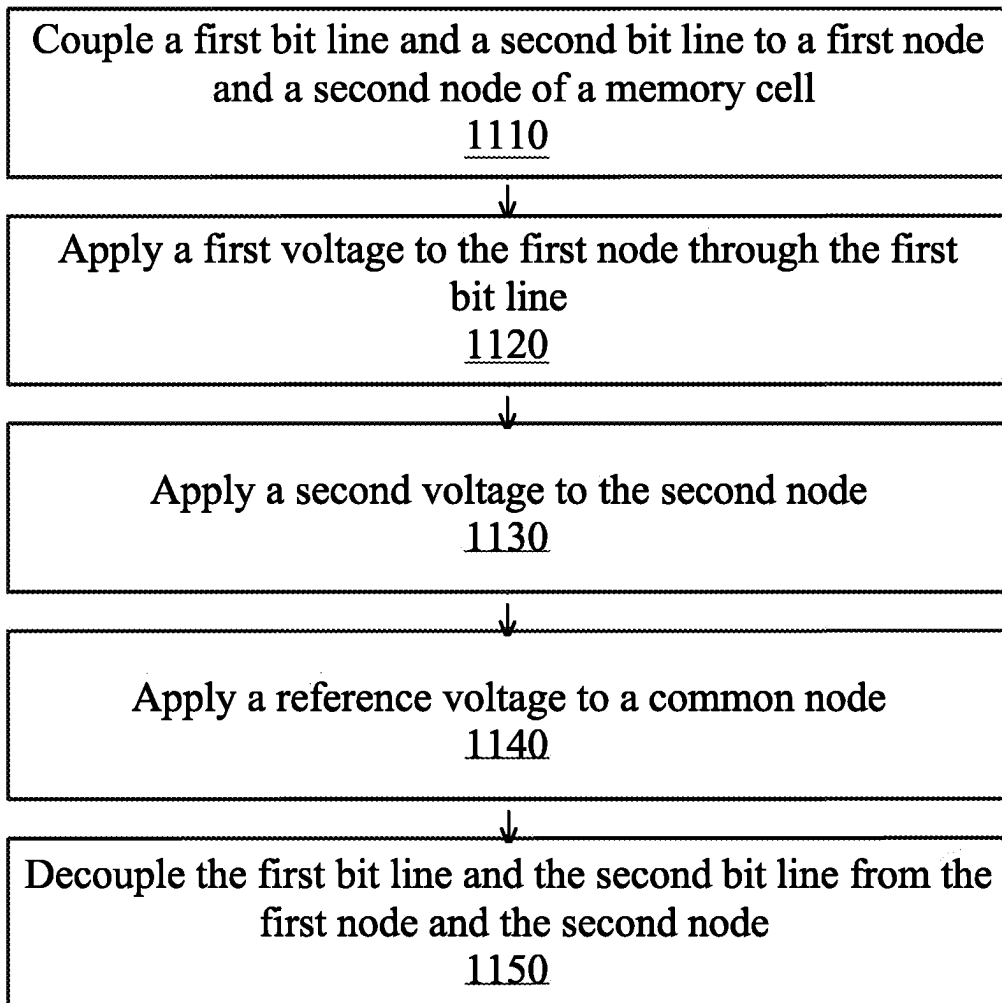
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FIG. 11

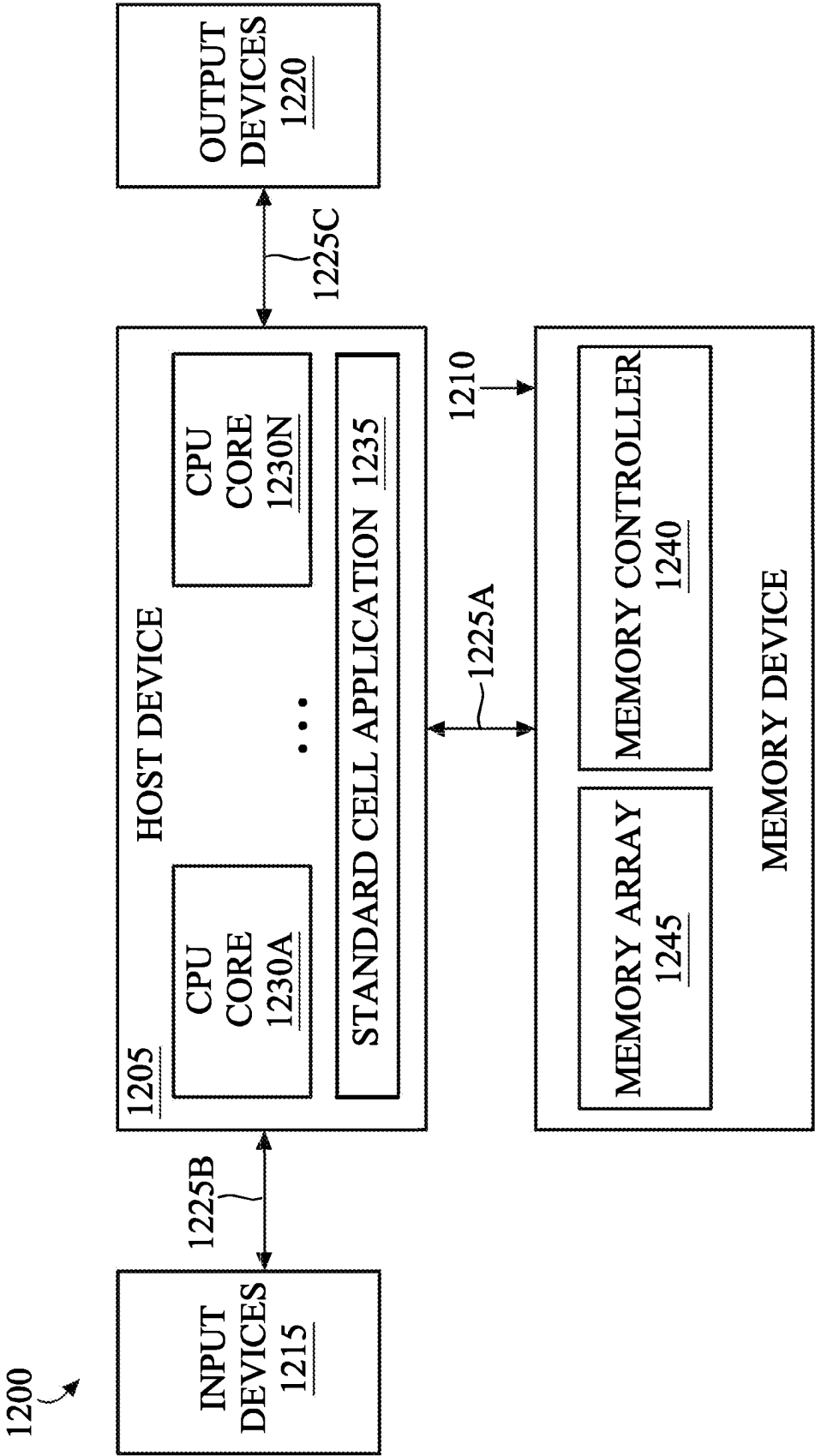


FIG. 12

STATIC RANDOM ACCESS MEMORY WITH MAGNETIC TUNNEL JUNCTION CELLS

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/853,206, filed Jun. 29, 2022, which is a continuation of U.S. patent application Ser. No. 17/166,570, filed Feb. 3, 2021, which claims priority to and the benefit of U.S. Provisional Application No. 63/016,420, filed Apr. 28, 2020, each of which are incorporated herein by reference in their entirety for all purposes.

BACKGROUND

Developments in electronic devices, such as computers, portable devices, smart phones, internet of thing (IoT) devices, etc., have prompted increased demands for memory devices. In general, memory devices may be volatile memory devices and non-volatile memory devices. Volatile memory devices can store data while power is provided, but may lose the stored data once the power is shut off. Unlike volatile memory devices, non-volatile memory devices may retain data even after the power is shut off but may be slower than the volatile memory devices.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a diagram of a memory system, in accordance with one embodiment.

FIG. 2 is a schematic diagram of a memory cell with magnetic tunnel junction (MTJ) devices, in accordance with some embodiments.

FIG. 3A is a diagram of an MTJ device in a parallel magnetic state, in accordance with some embodiments.

FIG. 3B is a diagram of the MTJ device in an anti-parallel magnetic state, in accordance with some embodiments.

FIG. 4 is a timing diagram showing an operation of the memory cell of FIG. 2, in accordance with some embodiments.

FIG. 5A is a schematic diagram of the memory cell of FIG. 2 storing a first state of a bit, in accordance with some embodiments.

FIG. 5B is a schematic diagram of the memory cell of FIG. 2 storing a second state of a bit, in accordance with some embodiments.

FIG. 6 is a top plan view of a memory cell with MTJ devices, in accordance with some embodiments.

FIG. 7 is a cross-section diagram along a line A-A' of the memory cell in FIG. 6, in accordance with some embodiments.

FIG. 8 is a top plan view of a memory cell with MTJ devices, in accordance with some embodiments.

FIG. 9 is a cross-section diagram along a line B-B' of the memory cell in FIG. 8, in accordance with some embodiments.

FIG. 10A is a top plan view of MTJ devices and power rails providing power to the MTJ devices, in accordance with some embodiments.

FIG. 10B is a top plan view of MTJ devices and power rails providing power to the MTJ devices, in accordance with some embodiments.

FIG. 11 is a flowchart of a method of writing data to a memory cell with MTJ devices, in accordance with some embodiments.

FIG. 12 is an example block diagram of a computing system, in accordance with some embodiments.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

In accordance with some embodiments, a storage circuit or a memory cell includes cross-coupled transistors and magnetic tunnel junction (MTJ) devices. In one aspect, the cross-coupled transistors are disposed on a first layer and the MTJ devices are disposed on a second layer above the first layer. The cross-coupled transistors may include a first transistor and a second transistor. The first transistor and the second transistor may be any transistors (e.g., metal oxide semiconductor field effect transistor (MOSFET), fin FET (FinFET), gate all around FET (GAAFET), etc.). The first transistor and the second transistor may be N-type transistors. A first drain structure of the first transistor may be electrically coupled to a first gate structure of the second transistor and a second drain structure of the second transistor may be electrically coupled to a second gate structure of the first transistor. In one aspect, a first MTJ device is electrically coupled to the first drain structure of the first transistor and a second MTJ device is electrically coupled to the second drain structure of the second transistor.

Advantageously, the disclosed storage circuit or memory cell provides several benefits. In one aspect, the disclosed storage circuit or memory cell bypasses or omits P-type transistors, but instead includes or implements N-type transistors with MTJ devices in different layers in a partially or wholly overlapping manner. Accordingly, area efficiency can be achieved by obviating P-type transistors that are larger than N-type transistors. Moreover, power efficiency can be achieved, because leakage current through P-type transistors

can be obviated. In addition, the MTJ devices can store bits or data without power, such that the disclosed storage circuit or memory cell can operate or function as a non-volatile memory cell.

Although various embodiments disclosed herein are described with respect to a memory cell including cross-coupled transistors and MTJ devices coupled to the cross-coupled transistors, different components may be implemented in some embodiments. For example, an amplifier or a regenerative circuit can replace the cross-coupled transistors, where different types of non-volatile memory devices or variable resistors can replace the MTJ devices.

FIG. 1 is a diagram of a memory system 100, in accordance with one embodiment. In some embodiments, the memory system 100 includes a memory controller 105 and a memory array 120. The memory array 120 may include a plurality of storage circuits or memory cells 125 arranged in two- or three-dimensional arrays. Each memory cell 125 may be coupled to a corresponding word line WL and a corresponding bit line BL. The memory controller 105 may write data to or read data from the memory array 120 according to electrical signals through word lines WL and bit lines BL. In other embodiments, the memory system 100 includes more, fewer, or different components than shown in FIG. 1.

The memory array 120 is a hardware component that stores data. In one aspect, the memory array 120 is embodied as a semiconductor memory device. The memory array 120 includes a plurality of storage circuits or memory cells 125. The memory array 120 includes word lines WL0, WL1 . . . WLJ, each extending in a first direction (e.g., X-direction) and bit lines BL0, BL1 . . . BLK, each extending in a second direction (e.g., Y-direction). The word lines WL and the bit lines BL may be conductive metals or conductive rails. In one aspect, each memory cell 125 is coupled to a corresponding word line WL and a corresponding bit line, and can be operated according to voltages or currents through the corresponding word line WL and the corresponding bit line BL. In one aspect, each memory cell 125 includes cross-coupled transistors and MTJ devices. Each memory cell 125 may be Static Random Access Memory (SRAM) memory cell with MTJ devices. By employing cross-coupled transistors and MTJ devices as disclosed herein, storage density and power efficiency can be improved. In some embodiments, the memory array 120 includes additional lines (e.g., select lines, reference lines, reference control lines, power rails, etc.). Detailed descriptions on configurations and operations of the memory system 100 are provided below with respect to FIGS. 2 through 10.

The memory controller 105 is a hardware component that controls operations of the memory array 120. In some embodiments, the memory controller 105 includes a bit line controller 112, a word line controller 114, and a timing controller 110. In one configuration, the word line controller 114 is a circuit that provides a voltage or a current through one or more word lines WL of the memory array 120, and the bit line controller 112 is a circuit that provides or senses a voltage or current through one or more bit lines BL of the memory array 120. In one configuration, the timing controller 110 is a circuit that provides control signals or clock signals to synchronize operations of the bit line controller 112 and the word line controller 114. The bit line controller 112 may be coupled to bit lines BL of the memory array 120, and the word line controller 114 may be coupled to word lines WL of the memory array 120. In one example, to write data to a memory cell 125, the word line controller 114 provides a voltage or current to the memory cell 125 through

a word line WL coupled to the memory cell 125, and the bit line controller 112 applies a bias voltage to the memory cell 125 through a bit line BL coupled to the memory cell 125. In one example, to read data from a memory cell 125, the word line controller 114 provides a voltage or current to the memory cell 125 through a word line WL coupled to the memory cell 125, and the bit line controller 112 senses a voltage or current corresponding to data stored by the memory cell 125 through a bit line BL coupled to the memory cell 125. In some embodiments, the memory controller 105 includes more, fewer, or different components than shown in FIG. 1.

FIG. 2 is a diagram of a memory cell 125 with MTJ devices 220A, 220B, in accordance with some embodiments. In some embodiments, the memory cell 125 includes MTJ devices 220A, 220B, and transistors T1, T2, T3, T4. These components may operate together to store a bit. In other embodiments, the memory cell 125 includes more, fewer, or different components than shown in FIG. 2.

In some embodiments, the transistors T1, T2 operate as a regenerative circuit or a positive feedback circuit. The transistor T1 and the transistor T2 may be any transistors (e.g., MOSFETs, FinFETs, GAAFETs, etc.). The transistor T1 and the transistor T2 may be N-type transistors. In one aspect, the transistors T1, T2 embodied as N-type transistors function or operate as pull-down transistors. In some embodiments, the transistors T1, T2 can be replaced by other circuits or components that perform the functionalities of the transistors T1, T2 described herein. In one configuration, the transistors T1, T2 are arranged in a cross-coupled configuration. In one aspect, a gate structure of the transistor T1 is coupled to a node N2, or a drain structure of the transistor T2. In one aspect, a gate structure of the transistor T2 is coupled to a node N1 or a drain structure of the transistor T1. In one aspect, a source structure of the transistor T1 and a source structure of the transistor T2 are coupled to each other. For example, the source structure of the transistor T1 and the source structure of the transistor T2 are coupled to a ground rail, at which a ground voltage is supplied. In this configuration, the cross-coupled transistors T1, T2 can sense a voltage difference of the voltages at the nodes N1, N2, and increase the voltage difference. For example, if a voltage at the node N1 is 0.498 V and a voltage at the node N2 is 0.497 V, the cross-coupled transistors T1, T2 can sense that the voltage at the node N1 is higher than the voltage at the node N2. In response to a higher voltage being applied to the gate structure of the transistor T2 than a voltage applied to the gate structure of the transistor T1, the transistor T2 may conduct more current than the transistor T1. By conducting more current through the transistor T2, the voltage at the node N2 may decrease to increase a voltage difference of the voltages at the nodes N1, N2. Through positive feedback, the voltage difference of the voltages at the nodes N1, N2 may increase, until one of the voltages at the nodes N1, N2 becomes, for example, the ground voltage.

In one configuration, the transistors T3, T4 operate as electrical switches or pass-gate transistors. The transistor T3 and the transistor T4 may be any transistors (e.g., MOSFETs, FinFETs, GAAFETs, etc.). The transistor T3 and the transistor T4 may be N-type transistors. In some embodiments, the transistors T3, T4 can be replaced by other circuits or components that perform the functionalities of the transistors T3, T4 described herein. In one configuration, the transistor T3 includes a gate structure electrically coupled to a word line WL, a source structure electrically coupled to a bit line BL, and a drain structure electrically coupled to the node N1 or the drain structure of the transistor T1. In one

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configuration, the transistor **T4** includes a gate structure electrically coupled to the word line **WL**, a source structure electrically coupled to a bit line **BLB**, and a drain structure electrically coupled to the node **N2** or the drain structure of the transistor **T2**. In this configuration, the transistor **T3** may allow the bit line **BL** to electrically couple to or decouple from the node **N1**, according to a voltage applied to the word line **WL**. Similarly, the transistor **T4** may allow the bit line **BLB** to electrically couple to or decouple from the node **N2**, according to the voltage applied to the word line **WL**. For example, according to a high voltage (e.g., **VDD**) applied to the word line **WL**, the transistor **T3** is enabled to electrically couple the bit line **BL** to the node **N1** and the transistor **T4** is enabled to electrically couple the bit line **BLB** to the node **N2**. For another example, according to a low voltage (e.g., **GND**) applied to the word line **WL**, the transistor **T3** is disabled to electrically decouple the bit line **BL** from the node **N1** and the transistor **T4** is disabled to electrically decouple the bit line **BLB** from the node **N2**.

The MTJ devices **220A**, **220B** are circuits that store a bit of data. The MTJ devices **220A**, **220B** may be embodied as non-volatile memory devices. The MTJ devices **220A**, **220B** may operate as variable resistors. In some embodiments, MTJ devices **220A**, **220B** can be replaced by other circuits or components that perform the functionalities of the MTJ devices **220A**, **220B** described herein. In one aspect, a resistance of the MTJ device **220** is adjusted or modified according to a voltage applied across the MTJ device **220**. In one configuration, the MTJ device **220** is electrically coupled between a drain structure of a transistor and a common node **N3**, at which a reference voltage (e.g., $\frac{1}{2}$ **VDD** or **0.5V**) is applied. The common node **N3** may be electrically coupled to or embodied as a metal rail or a power line providing a reference voltage. For example, the MTJ device **220A** is electrically coupled between the drain structure of the transistor **T1** and the common node **N3**, and the MTJ device **220B** is electrically coupled between the drain structure of the transistor **T2** and the common node **N3**. In this configuration, the MTJ devices **220A**, **220B** can store data according to voltages at the nodes **N1**, **N2**. For example, if a low voltage (e.g., **GND**) is applied to the node **N1** and a high voltage (e.g., **VDD**) is applied to the node **N2** while a reference voltage (e.g., $\frac{1}{2}$ **VDD** or **0.5V**) is applied to the node **N3**, the MTJ device **220A** may be set or programmed to have a lower resistance than a resistance of the MTJ device **220B**. For example, if the high voltage (e.g., **VDD**) is applied to the node **N1** and the low voltage (e.g., **GND**) is applied to the node **N2** while the reference voltage (e.g., $\frac{1}{2}$ **VDD** or **0.5V**) is applied to the node **N3**, the MTJ device **220A** may be set or programmed to have a higher resistance than a resistance of the MTJ device **220B**. Detailed descriptions on example operations of the MTJ device **220** are provided below with respect to FIGS. **3A** and **3B**.

In some embodiments, the memory controller **105** may provide signals (e.g., voltage and/or current) through the word line **WL** and the bit lines **BL**, **BLB** to write data to or read data from the memory cell **125**. When writing data, the memory controller **105** may enable the transistors **T3**, **T4** by providing a high voltage (e.g., **VDD**) to the word line **WL**. Then, the memory controller **105** may provide voltages corresponding to data to write to the nodes **N1**, **N2** through the bit lines **BL**, **BLB** and the transistors **T3**, **T4**. According to the voltages applied through the bit lines **BL**, **BLB**, the MTJ devices **220A**, **220B** can be set or programmed. For example, by applying the low voltage (e.g., **GND**) to the node **N1** and a reference voltage (e.g., $\frac{1}{2}$ **VDD**) to the node **N3** for a predetermined time period (e.g., **30 ns**), the MTJ

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device **220A** may have a parallel state **R_p** with a low resistance (e.g., less than **40 kΩ**). For example, by applying the high voltage (e.g., **VDD**) to the node **N2** and the reference voltage to the node **N3**, the MTJ device **220B** may have an anti-parallel state **R_{ap}** with a high resistance (e.g., larger than **80 kΩ**). After enabling the transistors **T3**, **T4** for the predetermined time period (e.g., **30 ns**) to set the states of the MTJ devices **220A**, **220B**, the memory controller **105** may disable the transistors **T3**, **T4** to decouple the bit lines **BL**, **BLB** from the nodes **N1**, **N2**, respectively.

When reading data, the memory controller **105** may disable the transistors **T3**, **T4** and provide a supply voltage (e.g., **VDD**) to the node **N3**. When the transistors **T3**, **T4** are disabled, voltages at the nodes **N1**, **N2** may be set, according to the programmed states of the MTJ devices **220A**, **220B**. For example, if the MTJ device **220A** has a lower resistance than the MTJ device **220B**, the voltage at the node **N1** can be closer to the supply voltage (e.g., **VDD**) than the voltage at the node **N2** where the voltage at the node **N2** can be closer to the ground voltage (e.g., **GND**) than the voltage at the node **N1**. For example, if the MTJ device **220B** has a lower resistance than the MTJ device **220A**, the voltage at the node **N2** can be closer to the supply voltage (e.g., **VDD**) than the voltage at the node **N1** where the voltage at the node **N1** can be closer to the ground voltage (e.g., **GND**) than the voltage at the node **N2**. After a predetermined time period (e.g., **10 ns**) for the voltages at the nodes **N1**, **N2** to settle, the memory controller **105** may enable the transistors **T3**, **T4** and sense voltages at the nodes **N1**, **N2** through the bit lines **BL**, **BLB**, respectively for reading. According to the sensed voltages through the bit lines **BL**, **BLB**, the memory controller **105** may determine a programmed bit of the memory cell **125**. For example, in response to the voltage sensed through the bit line **BL** being higher than the voltage sensed through the bit line **BLB**, the memory controller **105** may determine that the memory cell **125** stores a logic value '1'. For example, in response to the voltage sensed through the bit line **BL** being lower than the voltage sensed through the bit line **BLB**, the memory controller **105** may determine that the memory cell **125** stores a logic value '0'.

Advantageously, the memory cell **125** provides several benefits. In one aspect, the memory cell **125** bypasses or omits P-type transistors, but instead includes or implements N-type transistors **T1**, **T2**, **T3**, **T4** with MTJ devices **220A**, **220B**. The MTJ devices **220**, **220B** may be disposed in different layers than the transistors **N1**, **N2**, **N3**, **N4**. Accordingly, area efficiency can be achieved by obviating P-type transistors that are larger than N-type transistors. Moreover, power efficiency can be achieved, because leakage current through P-type transistors can be obviated. In addition, the MTJ devices can store bits or data without power, such that the memory cell **125** can operate or function as a non-volatile memory cell.

FIG. **3A** is a diagram of an MTJ device **220** in a parallel magnetic state, in accordance with some embodiments. FIG. **3B** is a diagram of the MTJ device **220** in an anti-parallel magnetic state, in accordance with some embodiments.

In some embodiments, the MTJ device **220** includes a free layer structure **310**, a barrier layer structure **315**, and a pinned layer structure **320**. The barrier layer structure **315** may separate the free layer structure **310** and the pinned layer structure **320**. In one aspect, a state of the pinned layer structure **320** is fixed or not configurable, where a state of the free layer structure **310** is configurable according to a voltage across the MTJ device **220**. For example, if a reference voltage (e.g., $\frac{1}{2}$ **VDD** or **0.5V**) is applied to the free layer structure **310** of the MTJ device **220** and a low

voltage (e.g., GND) lower than the reference voltage is applied to the pinned layer structure **320** of the MTJ device **220** for at least a time period (e.g., 30 ns), the MTJ device **220** can be programmed to have a parallel state R_p . For example, if the reference voltage (e.g., $\frac{1}{2}$ VDD or 0.5V) is applied to the free layer structure **310** structure of the MTJ device **220** and the high voltage (e.g., VDD) higher than the reference voltage is applied to the pinned layer structure **320** of the MTJ device **220** for at least a time period (e.g., 30 ns), the MTJ device **220** can be programmed to have an anti-parallel state R_p . The MTJ device **220** having the parallel state R_p may have a lower resistance than in the anti-parallel state R_p .

FIG. 4 is a timing diagram **400** showing an operation of the memory cell **125** of FIG. 2, in accordance with some embodiments. In one example, between time t_0 and time t_1 , the memory controller **105** may enable the transistors **T3**, **T4** to write data to the memory cell **125**. In one example, the memory controller **105** may enable the transistors **T3**, **T4** by providing a high voltage (e.g., VDD) to the word line **WL**. Then, the memory controller **105** may provide the low voltage (e.g., GND) to the node **N1** and a reference voltage V_{ref} (e.g., $\frac{1}{2}$ VDD) to the node **N3** for a predetermined time period (e.g., 30 ns). According to the low voltage (e.g., GND) provided to the node **N1** and the reference voltage V_{ref} applied to the node **N3** for a predetermined time period, the MTJ device **220A** may have a parallel state R_p with a low resistance (e.g., less than 40 k Ω). Meanwhile, the memory controller **105** may provide the high voltage (e.g., VDD) to the node **N2** and the reference voltage V_{ref} to the node **N3**. According to the high voltage (e.g., VDD) provided to the node **N2** and the reference voltage V_{ref} provided to the node **N3** for the predetermined time period, the MTJ device **220B** may have an anti-parallel state R_p with a high resistance (e.g., less than 80 k Ω). At time t_1 , after setting or programming the states of the MTJ devices **220A**, **220B**, the memory controller **105** may disable the transistors **T3**, **T4** to decouple the bit lines **BL**, **BLB** from the nodes **N1**, **N2**, respectively. At time t_1 , the memory controller **105** may discontinue providing a reference voltage to the node **N3**, such that the memory cell **125** may not consume power. By disabling the transistors **T3**, **T4** and stop providing the reference voltage to the node **N3**, voltages at the nodes **N1**, **N2** may become the ground voltage.

At time t_2 , power may be provided to the node **N3** to read data stored by the MTJ devices **220A**, **220B**. By providing a supply voltage (e.g., VDD) to the node **N3** while the transistors **T3**, **T4** are disabled, voltages at the nodes **N1**, **N2** may be set, according to the programmed states of the MTJ devices **220A**, **220B**. For example, if the MTJ device **220A** has a lower resistance than the MTJ device **220B**, the voltage at the node **N1** can be closer to the supply voltage (e.g., VDD) than the voltage at the node **N2** where the voltage at the node **N2** can be closer to the ground voltage (e.g., GND) than the voltage at the node **N1**. For example, if the MTJ device **220B** has a lower resistance than the MTJ device **220A**, the voltage at the node **N2** can be closer to the supply voltage (e.g., VDD) than the voltage at the node **N1** where the voltage at the node **N1** can be closer to ground voltage (e.g., GND) than the voltage at the node **N2**.

At time t_3 after the voltages at the nodes **N1**, **N2** settle according to the programmed states of the MTJ devices **220A**, **220B**, the memory controller **105** may enable the transistors **T3**, **T4** to sense voltages at the nodes **N1**, **N2** through the bit lines **BL**, **BLB**, respectively for reading. According to the sensed voltages through the bit lines **BL**, **BLB**, the memory controller **105** may determine a pro-

grammed bit of the memory cell **125**. For example, in response to the voltage sensed through the bit line **BL** being higher than the voltage sensed through the bit line **BLB**, the memory controller **105** may determine that the memory cell **125** stores a logic value '1'. For example, in response to the voltage sensed through the bit line **BL** being lower than the voltage sensed through bit line **BLB**, the memory controller **105** may determine that the memory cell **125** stores a logic value '0'.

FIG. 5A is a schematic diagram of the memory cell **125** storing a first state (e.g., logic value '1') of a bit, in accordance with some embodiments. The memory controller **105** may provide the low voltage (e.g., GND) to the node **N1** and a reference voltage (e.g., $\frac{1}{2}$ VDD) to the node **N3** for a predetermined time period (e.g., 30 ns). According to the low voltage (e.g., GND) provided to the node **N1** and the reference voltage (e.g., $\frac{1}{2}$ VDD) applied to the node **N3** for the predetermined time period, the MTJ device **220A** may have a parallel state R_p with a low resistance. Meanwhile, the memory controller **105** may provide the high voltage (e.g., VDD) to the node **N2** and the reference voltage (e.g., $\frac{1}{2}$ VDD) to the node **N3**. According to the high voltage (e.g., VDD) provided to the node **N2** and the reference voltage (e.g., $\frac{1}{2}$ VDD) provided to the node **N3** for the predetermined time period, the MTJ device **220B** may have an anti-parallel state R_p with a high resistance. After setting or programming the states of the MTJ devices **220A**, **220B** by applying the voltages for a predetermined time period (e.g., 30 ns). For reading data, transistors **T3**, **T4** may be disabled and a supply voltage (e.g., VDD) may be provided to the node **N3**. By providing the supply voltage and disabling the transistors **T3**, **T4**, voltages at the nodes **N1**, **N2** may settle, according to the programmed states of the MTJ devices **220A**, **220B**. In the example shown in FIG. 5A, the voltage at the node **N1** can be closer to the supply voltage (e.g., VDD) than the voltage at the node **N2** and the voltage at the node **N2** can be closer to the ground voltage (e.g., GND) than the voltage at the node **N1**, because the MTJ device **220A** has a lower resistance than the MTJ device **220B**. After the voltages at the nodes **N1**, **N2** settle, the memory controller **105** may enable the transistors **T3**, **T4** and sense voltages at the nodes **N1**, **N2** through the bit lines **BL**, **BLB**, respectively for reading. In the example shown in FIG. 5A, the memory controller **105** may determine that the memory cell **125** stores a logic value '1', because the voltage sensed through the bit line **BL** is higher than the voltage sensed through the bit line **BLB**.

FIG. 5B is a schematic diagram of the memory cell **125** of FIG. 2 storing a second state (e.g., logic value '0') of a bit, in accordance with some embodiments. The memory controller **105** may provide the high voltage (e.g., VDD) to the node **N1** and a reference voltage (e.g., $\frac{1}{2}$ VDD) to the node **N3** for a predetermined time period (e.g., 30 ns). According to the high voltage (e.g., VDD) provided to the node **N1** and the reference voltage (e.g., $\frac{1}{2}$ VDD) applied to the node **N3** for the predetermined time period, the MTJ device **220A** may have an anti-parallel state R_p with a high resistance. Meanwhile, the memory controller **105** may provide the low voltage (e.g., GND) to the node **N2** and the reference voltage (e.g., $\frac{1}{2}$ VDD) to the node **N3**. According to the low voltage (e.g., GND) provided to the node **N2** and the reference voltage (e.g., $\frac{1}{2}$ VDD) provided to the node **N3** for the predetermined time period, the MTJ device **220B** may have a parallel state R_p with a low resistance. After setting or programming the states of the MTJ devices **220A**, **220B** by applying the voltages for a predetermined time period (e.g., 30 ns). For reading data, transistors **T3**, **T4** may be disabled

and a supply voltage (e.g., VDD) may be provided to the node N3. By providing the supply voltage and disabling the transistors T3, T4, voltages at the nodes N1, N2 may settle, according to the programmed states of the MTJ devices 220A, 220B. In the example shown in FIG. 5B, the voltage at the node N2 can be closer to the supply voltage (e.g., VDD) than the voltage at the node N1 and the voltage at the node N1 can be closer to the ground voltage (e.g., GND) than the voltage at the node N2, because the MTJ device 220B has a lower resistance than the MTJ device 220A. After the voltages at the nodes N1, N2 settle, the memory controller 105 may enable the transistors T3, T4 and sense voltages at the nodes N1, N2 through the bit lines BL, BLB, respectively for reading. In the example shown in FIG. 5B, the memory controller 105 may determine that the memory cell 125 stores a logic value '0', because the voltage sensed through the bit line BLB is higher than the voltage sensed through the bit line BL.

FIG. 6 is a layout or a top plan view 600 of the memory cell 125 with MTJ devices 220A, 220B, in accordance with some embodiments. In some embodiments, the memory cell 125 includes gate structures 610A, 610B, 610C, 610D elongated along a X-direction, active regions 630A, 630B, 630C, 630D elongated along a Y-direction, and MTJ devices 220A, 220B. These components may be arranged and function as the memory cell 125 described above with respect to FIG. 6. In one aspect, the memory cell 125 includes more, fewer, or different components than shown in FIG. 6. For example, the memory cell 125 includes additional components (e.g., routing metals, via contacts) that are not shown in FIG. 6.

In some embodiments, transistors are formed where the gate structures 610A, 610B, 610C, 610D and active regions 630A, 630B, 630C, 630D intersect. The gate structures 610A, 610B, 610C, 610D may correspond to or include polysilicon or other conductive materials. The active regions 630A, 630B, 630C, 630D may include N-diffusion. For example, a transistor T1 is formed, at which the active regions 630A, 630B and the gate structure 610B intersect. For example, a transistor T3 is formed, at which the active regions 630A, 630B and the gate structure 610A intersect. For example, a transistor T2 is formed, at which the active regions 630C, 630D and the gate structure 610C intersect. For example, a transistor T4 is formed, at which the active regions 630C, 630D and the gate structure 610D intersect. In one example, the transistors T1 and T3 share drain structures in the active region 630A, 630B underneath the MTJ device 220A. In one example, the transistors T2 and T4 share drain structures in the active region 630C, 630D underneath the MTJ device 220B.

In one aspect, the MTJ devices 220A, 220B are disposed on a different layer than a layer of the transistors T1, T2, T3, T4. In one aspect, the MTJ devices 220A, 220B may partially overlap with the transistors T1, T2, T3, T4. For example, the MTJ device 220A is disposed above, or overlaps with the shared drain structure of the transistors T1, T3. For example, the MTJ device 220B is disposed above, or overlaps with the shared drain structure of the transistors T2, T4.

Advantageously, the memory cell 125 can be formed in an area efficient manner. In one aspect, by sharing the drain structures, the transistors T1, T2, T3, T4 can be formed in a compact area. Moreover, by obviating P-type transistors that generally consume larger areas than N-type transistors, the memory cell 125 can be formed in a smaller area. Furthermore, by forming the MTJ devices 220A, 220B above the

shared drain structures of the transistors T1, T2, T3, T4, the transistors T1, T2, T3, T4 can be formed in a compact area to achieve area efficiency.

FIG. 7 is a cross-section diagram along a line A-A' of the memory cell 125 in FIG. 6, in accordance with some embodiments. In one aspect, the active region 630A extends along the Y-direction. In a first layer 620, the transistor T3 may be formed where the gate structure 610A and the active region 630A intersect or overlap, and the transistor T1 may be formed where the gate structure 610B and the active region 630A intersect or overlap. As shown in FIG. 7, the transistors T1, T3 may share the drain structure D. The shared drain structure D may be or function as the node N1. In a second layer 680 above the first layer 620, the MTJ device 220A is formed above the shared drain structure D of the transistors T1, T3 along the Z-direction. In one configuration, the pinned layer structure 320 is disposed in a lower layer than the free layer structure 310. Hence, the pinned layer structure 320 of the MTJ device 220A can be electrically coupled to the drain structure D through via contacts (e.g., VD . . . Vx-1) and metal rails (M0 . . . Mx), and the free layer structure 310 of the MTJ device 220A can be electrically coupled to a metal rail Mx+1 through a via contact Vx. The metal rail Mx+1 may be or function as the node N3.

FIG. 8 is a top plan 800 view of a memory cell 125 with MTJ devices 220A, 220B, in accordance with some embodiments. FIG. 9 is a cross-section diagram along a line B-B' of the memory cell 125 in FIG. 8, in accordance with some embodiments. In one aspect, the memory cell 125 shown in FIG. 8 is similar to the memory cell 125 shown in FIG. 6, except the MTJ devices 220A, 220B are disposed along a diagonal line traversing the X direction and the Y direction. In some embodiments, the MTJ devices 220A, 220B are disposed in metal 3 layer or a higher layer. A pinned layer structure of the MTJ device 220 may be electrically coupled to a drain structure D of a transistor through an interconnect 910 (e.g., metal 0 layer, metal 1 layer, or metal 2 layer) below the MTJ devices 220A, 220B. The interconnect 910 may extend along the Y-direction to overlap with the drain structure of the transistor T1 and the gate structure 610B of the transistor T1. In one example, the drain structure D of the transistor T1 is electrically connected to a first part of the interconnect 910 through a metal diffusion (MD), a via contact VD, a metal rail M0, and a via contact V0. In one example, a second part of the interconnect 910 is electrically connected to a pinned layer structure of the MTJ device 220A through the metal rails and via contacts between the interconnect 910 and the MTJ device 220A. By implementing the interconnect 910, the MTJ device 220A may be disposed above the gate structure 610B of the transistor T1 and the MTJ device 220B may be disposed above the gate structure 610C of the transistor T2.

Advantageously, a memory cell can be implemented in a compact manner while satisfying spacing rules or requirements (e.g., design rule checking (DRC)) by placing the MTJ devices 220A, 220B along the diagonal direction as shown in FIGS. 8 and 9. In one example, MTJ devices 220A, 220B separated by a distance less than a separation distance along the X-direction or the Y-direction may violate DRC. A separation distance may be 1.5~2 times the distance between two active regions 630. By implementing the MTJ devices 220A, 220B along the diagonal direction as shown in FIGS. 8 and 9, the MTJ devices 220A, 220B may be implemented within a rectangular area allocated for placing four transistors T1, T2, T3, T4, while satisfying the spacing rules or requirements.

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In some embodiments, each MTJ device **220** has a width *W* along the X-direction and a length *L* along the Y-direction. The length *L* of each MTJ device **220** may be larger than a polypitch CPP but less than 1.5 times the polypitch CPP. A polypitch CPP may be a distance between a center of a first gate structure and a center of a second gate structure. By ensuring that the length *L* of the MTJ device **220** is less than 1.5 times the polypitch CPP, the diagonally located MTJ devices **220A**, **220B** may satisfy the spacing rules or requirements of the MTJ devices **220A**, **220B**. For example, by ensuring that the length *L* of each MTJ device **220A**/**220B** is less than 1.5 times the polypitch CPP, a right end of the MTJ device **220A** along the X-direction may not face a left end of the MTJ device **220B** along the X-direction. Hence, the MTJ devices **220A**, **220B** can be disposed along the diagonal direction to achieve area efficiency.

FIG. **10A** is a top plan view **1000A** of MTJ devices **220A**, **220B** and power rails **1010A**, **1010B** providing a reference voltage or a supply voltage to the MTJ devices **220A**, **220B**, in accordance with some embodiments. FIG. **10B** is a top plan view **1000B** of MTJ devices **220A**, **220B** and power rails **1010C**, **1010D** providing a reference voltage or a supply voltage to the MTJ devices **220A**, **220B**, in accordance with some embodiments. The power rails **1010** may constitute or function as the common node **N3**. In some embodiments, the power rails **1010A**, **1010B** may extend in the Y-direction as shown in FIG. **10A**. The power rail **1010A** may be electrically connected to a free layer structure of the MTJ device **220A**, and the power rail **1010B** may be electrically connected to a free layer structure of the MTJ device **220B**. In some embodiments, the power rails **1010C**, **1010D** may extend in the X-direction as shown in FIG. **10B**. In one aspect, the power rails **1010** may extend in the direction associated with a particular layer. For example, the power rails **1010** may extend in the X-direction for an even numbered layer (e.g., **M4**, **M6**, **M8**) or extend in the Y-direction for odd numbered layer (e.g., **M3**, **M5**, **M7**). In some embodiments, the metal rails **1010A**, **1010B** may be replaced by a single metal rail having a width sufficient to cover or overlap the MTJ devices **220A**, **220B**, such that a reference voltage or a supply voltage can be provided to the free layer structures of the MTJ devices **220A**, **220B** with low resistance.

FIG. **11** is a flowchart of a method **1100** of writing data to the memory cell **125** with MTJ devices **220A**, **220B**, in accordance with some embodiments. The method **1100** may be performed by the memory controller **105** of FIG. **1**. In some embodiments, the method **1100** is performed by other entities. In some embodiments, the method **1100** includes more, fewer, or different operations than shown in FIG. **11**.

In an operation **1110**, the memory controller **105** configures the transistors **T3**, **T4** to couple the first bit line **BL** to a first node **N1** and couple the second bit line **BLB** to a second node **N2**. In one approach, the memory controller **105** generates and applies a high voltage (e.g., **VDD**) to a word line **WL** coupled to the gate structures of the transistors **T3**, **T4** to enable the transistors **T3**, **T4**.

In an operation **1120**, the memory controller **105** applies a first voltage to the first node **N1** through the first bit line. In an operation **1130**, the memory controller **105** applies a second voltage to the second node **N2** through the second bit line. In an operation **1140**, the memory controller **105** applies a reference voltage to the third node **N3**. In one approach, the memory controller **105** applies a low voltage (e.g., **GND**) to the bit line **BL** while applying the high voltage (e.g., **VDD**) to the bit line **BLB** and the reference voltage (e.g., $\frac{1}{2}$ **VDD**) to the node **N3** to store a logic value

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'1'. In one approach, the memory controller **105** applies a high voltage (e.g., **VDD**) to the bit line **BL** while applying the low voltage (e.g., **GND**) to the bit line **BLB** and the reference voltage (e.g., $\frac{1}{2}$ **VDD**) to the node **N3** to store a logic value '0'. After applying the voltages for the predetermined time period (e.g., 30 ns), the states of the MTJ devices **220A**, **220B** may be set or programmed. For example, if the low voltage (e.g., **GND**) is applied to the node **N1** and the high voltage (e.g., **VDD**) is applied to the node **N2** while the reference voltage (e.g., $\frac{1}{2}$ **VDD** or 0.5V) is applied to the node **N3**, the MTJ device **220A** may be set or programmed to have a parallel state **Rp** and the MTJ device **220B** may be set or programmed to have an anti-parallel state **Rap**. For example, if the high voltage (e.g., **VDD**) is applied to the node **N1** and the low voltage (e.g., **GND**) is applied to the node **N2** while the reference voltage (e.g., $\frac{1}{2}$ **VDD** or 0.5V) is applied to the node **N3**, the MTJ device **220A** may be set or programmed to have an anti-parallel state **Rap** and the MTJ device **220B** may be set or programmed to have a parallel state **Rp**.

In an operation **1150**, the memory controller **105** decouples the first bit line **BL** from the first node **N1** and the second bit line **BLB** from the second node **N2**. In one approach, the memory controller **105** generates and applies a low voltage (e.g., **GND**) to the word line **WL** coupled to the gate structures of the transistors **T3**, **T4** to disable the transistors **T3**, **T4**. The memory controller **105** may apply a low voltage (e.g., **GND**) to the common node **N3** to power off the memory cell **125**. In one aspect, the MTJ devices **220A**, **220B** may retain data without power.

To read data stored by the memory cell **125**, the memory controller **105** may apply a supply voltage (e.g., **VDD**) to the node **N3**, while the transistors **T3**, **T4** are disabled. By applying the supply voltage (e.g., **VDD**) to the node **N3**, voltages at the nodes **N1**, **N2** may be set, according to the programmed states of the MTJ devices **220A**, **220B**. For example, if the MTJ device **220A** has a lower resistance than the MTJ device **220B**, the voltage at the node **N1** can be closer to the supply voltage (e.g., **VDD**) than the voltage at the node **N2** where the voltage at the node **N2** can be closer to the ground voltage (e.g., **GND**) than the voltage at the node **N1**. For example, if the MTJ device **220B** has a lower resistance than the MTJ device **220A**, the voltage at the node **N2** can be closer to the supply voltage (e.g., **VDD**) than the voltage at the node **N1** where the voltage at the node **N1** can be closer to the ground voltage (e.g., **GND**) than the voltage at the node **N2**. After the voltages at the nodes **N1**, **N2** settle, the memory controller **105** may enable the transistors **T3**, **T4** to sense voltages at the nodes **N1**, **N2** and determine data stored by the memory cell **125** according to the sensed voltages.

Advantageously, the disclosed memory cell **125** provides several benefits. In one aspect, the memory cell **125** bypasses or omits P-type transistors, but instead includes or implements N-type transistors with MTJ devices **220** in different layers in a partially or wholly overlapping manner. Accordingly, area efficiency can be achieved by obviating P-type transistors that are larger than N-type transistors. Moreover, power efficiency can be achieved, because leakage current through P-type transistors can be obviated. In addition, the MTJ devices **220** can store bits or data without power, such that the memory cell **125** can operate or function as a non-volatile memory cell.

Referring now to FIG. **12**, an example block diagram of a computing system **1200** is shown, in accordance with some embodiments of the disclosure. The computing system **1200** may be used by a circuit or layout designer for integrated

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circuit design. A “circuit” as used herein is an interconnection of electrical components such as resistors, transistors, switches, batteries, inductors, or other types of semiconductor devices configured for implementing a desired functionality. The computing system **1200** includes a host device **1205** associated with a memory device **1210**. The host device **1205** may be configured to receive input from one or more input devices **1215** and provide output to one or more output devices **1220**. The host device **1205** may be configured to communicate with the memory device **1210**, the input devices **1215**, and the output devices **1220** via appropriate interfaces **1225A**, **1225B**, and **1225C**, respectively. The computing system **1200** may be implemented in a variety of computing devices such as computers (e.g., desktop, laptop, servers, data centers, etc.), tablets, personal digital assistants, mobile devices, other handheld or portable devices, or any other computing unit suitable for performing schematic design and/or layout design using the host device **1205**.

The input devices **1215** may include any of a variety of input technologies such as a keyboard, stylus, touch screen, mouse, track ball, keypad, microphone, voice recognition, motion recognition, remote controllers, input ports, one or more buttons, dials, joysticks, and any other input peripheral that is associated with the host device **1205** and that allows an external source, such as a user (e.g., a circuit or layout designer), to enter information (e.g., data) into the host device and send instructions to the host device. Similarly, the output devices **1220** may include a variety of output technologies such as external memories, printers, speakers, displays, microphones, light emitting diodes, headphones, video devices, and any other output peripherals that are configured to receive information (e.g., data) from the host device **1205**. The “data” that is either input into the host device **1205** and/or output from the host device may include any of a variety of textual data, circuit data, signal data, semiconductor device data, graphical data, combinations thereof, or other types of analog and/or digital data that is suitable for processing using the computing system **1200**.

The host device **1205** includes or is associated with one or more processing units/processors, such as Central Processing Unit (“CPU”) cores **1230A-1230N**. The CPU cores **1230A-1230N** may be implemented as an Application Specific Integrated Circuit (“ASIC”), Field Programmable Gate Array (“FPGA”), or any other type of processing unit. Each of the CPU cores **1230A-1230N** may be configured to execute instructions for running one or more applications of the host device **1205**. In some embodiments, the instructions and data to run the one or more applications may be stored within the memory device **1210**. The host device **1205** may also be configured to store the results of running the one or more applications within the memory device **1210**. Thus, the host device **1205** may be configured to request the memory device **1210** to perform a variety of operations. For example, the host device **1205** may request the memory device **1210** to read data, write data, update or delete data, and/or perform management or other operations. One such application that the host device **1205** may be configured to run may be a standard cell application **1235**. The standard cell application **1235** may be part of a computer aided design or electronic design automation software suite that may be used by a user of the host device **1205** to use, create, or modify a standard cell of a circuit. In some embodiments, the instructions to execute or run the standard cell application **1235** may be stored within the memory device **1210**. The standard cell application **1235** may be executed by one or more of the CPU cores **1230A-1230N** using the instructions associated

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with the standard cell application from the memory device **1210**. In one example, the standard cell application **1235** allows a user to utilize pre-generated schematic and/or layout designs of the memory system **100** or a portion of the memory system **100** to aid integrated circuit design. After the layout design of the integrated circuit is complete, multiples of the integrated circuit, for example, including the memory system **100** or a portion of the memory system **100** can be fabricated according to the layout design by a fabrication facility.

Referring still to FIG. **12**, the memory device **1210** includes a memory controller **1240** that is configured to read data from or write data to a memory array **1245**. The memory array **1245** may include a variety of volatile and/or non-volatile memories. For example, in some embodiments, the memory array **1245** may include NAND flash memory cores. In other embodiments, the memory array **1245** may include NOR flash memory cores, SRAM cores, Dynamic Random Access Memory (DRAM) cores, Magnetoresistive Random Access Memory (MRAM) cores, Phase Change Memory (PCM) cores, Resistive Random Access Memory (ReRAM) cores, 3D XPoint memory cores, ferroelectric random-access memory (FeRAM) cores, and other types of memory cores that are suitable for use within the memory array. The memories within the memory array **1245** may be individually and independently controlled by the memory controller **1240**. In other words, the memory controller **1240** may be configured to communicate with each memory within the memory array **1245** individually and independently. By communicating with the memory array **1245**, the memory controller **1240** may be configured to read data from or write data to the memory array in response to instructions received from the host device **1205**. Although shown as being part of the memory device **1210**, in some embodiments, the memory controller **1240** may be part of the host device **1205** or part of another component of the computing system **1200** and associated with the memory device. The memory controller **1240** may be implemented as a logic circuit in either software, hardware, firmware, or combination thereof to perform the functions described herein. For example, in some embodiments, the memory controller **1240** may be configured to retrieve the instructions associated with the standard cell application **1235** stored in the memory array **1245** of the memory device **1210** upon receiving a request from the host device **1205**.

It is to be understood that only some components of the computing system **1200** are shown and described in FIG. **12**. However, the computing system **1200** may include other components such as various batteries and power sources, networking interfaces, routers, switches, external memory systems, controllers, etc. Generally speaking, the computing system **1200** may include any of a variety of hardware, software, and/or firmware components that are needed or considered desirable in performing the functions described herein. Similarly, the host device **1205**, the input devices **1215**, the output devices **1220**, and the memory device **1210** including the memory controller **1240** and the memory array **1245** may include other hardware, software, and/or firmware components that are considered necessary or desirable in performing the functions described herein.

One aspect of this description relates to a memory cell. In some embodiments, the memory cell includes a first layer including a first transistor and a second transistor. In some embodiments, the first transistor and the second transistor are connected to each other in a cross-coupled configuration. In some embodiments, a first drain structure of the first transistor is electrically coupled to a first gate structure of the

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second transistor. In some embodiments, a second drain structure of the second transistor is electrically coupled to a second gate structure of the first transistor. In some embodiments, the memory cell includes a second layer including a first magnetic tunnel junction (MTJ) device electrically coupled to the first drain structure of the first transistor and a second MTJ device electrically coupled to the second drain structure of the second transistor. In some embodiments, the second layer is above the first layer.

One aspect of this description relates to a memory system. In some embodiments, the system includes a memory cell including cross-coupled transistors electrically coupled to a first node and a second node. In some embodiments, the memory cell includes a first magnetic tunnel junction (MTJ) device coupled between a common node and the first node. In some embodiments, the memory cell includes a second MTJ device coupled between the common node and the second node. In some embodiments, the first MTJ device is disposed above a part of the cross-coupled transistors. In some embodiments, the second MTJ device is disposed above another part of the cross-coupled transistors. In some embodiments, the system includes a memory controller coupled to the memory cell. In some embodiments, the memory controller programs the memory cell. In some embodiments, programming the memory cell includes applying a first voltage to the first node. In some embodiments, programming the memory cell includes applying a second voltage to the second node while the first voltage is applied to the first node. In some embodiments, programming the memory cell includes applying a reference voltage to the common node while the first voltage is applied to the first node and the second voltage is applied to the second node.

One aspect of this description relates to a SRAM cell. In some embodiments, the SRAM cell includes a first MTJ device comprising a free layer structure, a barrier layer structure, and a pinned layer structure. In some embodiments, the SRAM cell includes a first pass-gate transistor coupled between a bit line and the pinned layer structure of the first MTJ device. In some embodiments, the SRAM cell includes a first pull-down transistor coupled between a ground and the pinned layer structure of the first MTJ device. In some embodiments, the free layer structure of the first MTJ device is coupled to a power line.

One aspect of this description relates to a method of operating a memory cell. In some embodiments, the method includes applying, by a memory controller, a first voltage to a first pinned layer structure of a first magnetic tunnel junction (MTJ) device of the memory cell. In some embodiments, the method includes applying, by the memory controller, a second voltage to a second pinned layer structure of a second MTJ device of the memory cell, while the first voltage is applied to the first pinned layer structure of the first MTJ device. The first pinned layer structure and the second pinned layer structure may be coupled to the cross-coupled transistors. In some embodiments, the method includes applying, by the memory controller, a reference voltage to a first free layer structure of the first MTJ device and a second free layer structure of the second MTJ device, while the first voltage is applied to the first pinned layer structure of the first MTJ device and the second voltage is applied to the second pinned layer structure of the second MTJ device.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present

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disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A memory cell comprising:

a first transistor and a second transistor connected to each other in a cross-coupled configuration, wherein the memory cell does not include a P-type transistor;
a third transistor coupled to a first drain structure of the first transistor and coupled to a bit line of the memory cell; and

a first magnetic tunnel junction (MTJ) device electrically coupled to the first drain structure of the first transistor and a second MTJ device electrically coupled to a second drain structure of the second transistor, wherein the first MTJ device and the second MTJ device are each coupled to a common node, the first MTJ device includes a free layer structure electrically coupled to the common node, and voltage is applied across each of the first MTJ device and the second MTJ device via the common node.

2. The memory cell of claim 1, further comprising a first layer including the first transistor and the second transistor.

3. The memory cell of claim 2, further comprising a second layer including the first MTJ device and the second MTJ device.

4. The memory cell of claim 3, wherein the second layer is above the first layer.

5. The memory cell of claim 1, wherein:

the first drain structure of the first transistor is electrically coupled to a first gate structure of the second transistor, and

the second drain structure of the second transistor is electrically coupled to a second gate structure of the first transistor.

6. The memory cell of claim 1, wherein the first MTJ device includes a pinned layer structure electrically coupled to the first drain structure of the first transistor.

7. The memory cell of claim 1, further comprising a fourth transistor electrically coupled between a second bit line and the second drain structure of the second transistor.

8. The memory cell of claim 7, wherein the first transistor, the second transistor, the third transistor, and the fourth transistor are each defined on a first layer of the memory cell.

9. A memory cell comprising:

a first layer comprising a first transistor, a second transistor, and a third transistor, the first transistor and the second transistor connected to each other in a cross-coupled configuration, the first transistor and the third transistor sharing a first drain structure; and

a second layer comprising:

a first magnetic tunnel junction (MTJ) device coupled to the first drain structure of the first transistor, and a second MTJ device electrically coupled to a second drain structure of the second transistor, wherein voltage is applied across each of the first MTJ device and the second MTJ device via a common node, and the first MTJ device comprises a free layer structure electrically coupled to the common node.

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10. The memory cell of claim 9, wherein the first MTJ device is disposed above the first drain structure of the first transistor.

11. The memory cell of claim 9, wherein the first MTJ device is disposed above a first gate structure of the first transistor.

12. The memory cell of claim 10, wherein the third transistor is electrically coupled between a first bit line and the first drain structure.

13. The memory cell of claim 10, wherein the first transistor, the second transistor, and the third transistor are N-type transistors.

14. The memory cell of claim 9, wherein the first MTJ device is electrically coupled to the first drain structure via one or more contacts.

15. The memory cell of claim 9, wherein the first MTJ device includes a pinned layer structure electrically coupled to the first drain structure of the first transistor.

16. A method of operating a memory cell comprising:
applying, by a memory controller, a first voltage to a first magnetic tunnel junction (MTJ) device of the memory cell;

applying, by the memory controller, a second voltage to a second MTJ device of the memory cell while the first voltage is applied to the first MTJ device, the first MTJ device electrically coupled to a first drain structure of a first transistor and the second MTJ device electrically coupled to a second drain structure of a second tran-

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sistor, wherein the first transistor and the second transistor are cross-coupled, wherein the second voltage is applied to the second MTJ device via a third transistor electrically coupled to the second MTJ device at the second drain structure, and wherein the first MTJ device and the second MTJ device are electrically coupled to a common node; and

applying, by the memory controller, a reference voltage to the common node while the first voltage and the second voltage are applied.

17. The method of claim 16, wherein the second voltage is applied to the second MTJ device via a bit line coupled to the third transistor.

18. The memory cell of claim 6, wherein the pinned layer structure is a first pinned layer structure, and the second MTJ device includes a second pinned layer structure electrically coupled to the second drain structure of the second transistor.

19. The memory cell of claim 6, wherein the pinned layer structure is disposed closer to the first transistor than the free layer structure.

20. The memory cell of claim 15, wherein the pinned layer structure is a first pinned layer structure, and the second MTJ device includes a second pinned layer structure electrically coupled to the second drain structure of the second transistor.

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